Karan Thakur

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Summary

Computer Engineering student with hands-on experience building scalable, mission-critical distributed systems using C++, TypeScript, and AWS. Experienced collaborating across technical and business teams to design, develop, and deliver reliable services under tight timelines, with a strong foundation in data structures, algorithms, and OOP. Eager to learn and adopt new technologies, contribute to real-world products at global scale, and empower users through creative and principled software engineering.

Education

Thapar University **Aug. 2023 – May 2027**
*Bachelor of Engineering (B.E.) in Computer Engineering | CGPA: **8.1 / 10*** *Patiala, Punjab*

Experience

Full Stack Intern – EduSprint **Mar. 2026 – May 2026**
Tech: AWS (CloudFront, S3), Node.js, React *Bangalore*

- Partnered with product and infrastructure teams to scope content-delivery requirements, translating business needs into a token-based **AWS** CloudFront signed cookie solution that enforced information security policy and restricted premium video content strictly to authenticated, paying users.
- Improved API response time by ∼**25%** through database indexing and query optimization, reducing latency on critical endpoints and directly improving the **quality of experience** for end users of the platform.
- Reduced origin server load by ∼**35%** by offloading video delivery to CloudFront CDN, cutting buffering incidents and improving content availability — measurably improving the end-user streaming experience.

Projects

Parallel Attention Engine | *C++, OpenMP, CUDA, Python* | GitHub **Apr. 2026**

- Parallelized Multi-Head Attention with OpenMP and CUDA on top of a sequential C++ baseline.
- Achieved up to **15× speedup**, reducing Multi-Head Attention runtime from **76,268 ms to under 5,000 ms**.
- Optimized CUDA kernels and OpenMP threading strategies, with correctness verified at error < 1e-4.

Transcoder | *Node.js, AWS (S3, Lambda, ECS, CloudFront), Docker, Redis, Kafka* | GitHub | Live **Jan. 2026**

- Designed and delivered an event-driven HLS video transcoding pipeline (360p–1080p) over a **3-month timeline**, serving **10,000+ concurrent users** via CloudFront CDN; cut processing time **30–40%** with parallel Node.js worker threads and sustained <**3s** time-to-first-segment under peak load.

Zapier Clone | *Node.js, TypeScript, PostgreSQL, Kafka, Redis, Docker, Kubernetes* | GitHub **Sep. 2025**

- Engineered a workflow-automation platform across **5 microservices** on **Kubernetes**, handling ∼**500 concurrent webhooks** at <**200 ms** latency; enforced **zero event loss** via Transactional Outbox Pattern and **exactly-once** transaction processing via idempotent Kafka consumers with Redis deduplication.

Technical Skills

Databases & Infra.: Redis, PostgreSQL, AWS, Docker, Kubernetes, Kafka
Languages: C, C++, CUDA, TypeScript, JavaScript, Python, SQL, HTML, CSS, MATLAB
Frameworks: Node.js, Express.js, Next.js, React.js
Testing: Jest, Vitest, Supertest, Cypress
Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming, Artificial Intelligence, Computer Networks

Positions of Responsibility

Core Member – ACM Society **Aug. 2024 – Aug. 2025**
Thapar University *Patiala, Punjab*

- Directed technical initiatives for an ACM chapter of **160+ students**, fostering peer learning and active engagement.
- Delivered **2-hour DSA workshops**, mentoring **50+ students** and improving their problem-solving skills.

Class Representative **Aug. 2024 – Dec. 2024**
Thapar University *Patiala, Punjab*

- Represented **250+ students** as the primary liaison between students and faculty.
- Coordinated academic concerns and updates, enabling faster issue resolution and improved class coordination.

Achievements

- **Solved 700+ DSA Questions:** Strengthened C++ problem-solving skills through LeetCode and GeeksforGeeks.
- **Coding Contests:** Participated in **25+ contests**, improving speed, accuracy, and algorithmic thinking.
- **Cloud Deployment:** Built and deployed production-grade applications using AWS, Docker, and Kubernetes.